

Function Words and Content Words

Function words & stop words (vs. content words)

Function words, also known as “**junk**” words or **stop words**, comprise the set of words represented by *pronouns* (e.g., I, you, we, mine), *articles* (e.g., a, the), *conjunctions* (e.g., and, or), *prepositions* (e.g., after, in, to, on, with), and auxiliary verbs (e.g., can, would). According to Pennebaker: “In the English language there are fewer than 200 commonly used particles [function words], yet they account for over half of the words we use.” (Pennebaker et al. 2003: 570; “fewer than 400... function words” in Chung and Pennebaker 2007: 347).

Content words, on the other hand, are represented by nouns, verbs, and adjectives. Content words in English include over 100,000 words, compared to 200 or 400 function words.

Pennebaker recommends focusing one’s attention on function words rather than content words in analyzing text. These words can tell us a great deal about a speaker’s/writer’s style, about gender differences in style, and about the psychological state of speakers/writers. Kestemont (2014: 60) also recommends moving “away from a language’s low-frequency features to a language’s high-frequency features, which often tend to be function words.”

Stanford CoreNLP tags function words with the Penn treebank tags

(https://www.ling.upenn.edu/courses/Fall_2003/ling001/penn_treebank_pos.html):

Pronouns (POStag = PRP, PRP\$, WP, WP\$, WDT)

Prepositions (POStag = IN)

Articles (POStag = DT)

Conjunctions (POStag = CC, IN)

Auxiliary verbs (DEPREL = aux, auxpass)

Pronouns

Pronouns can be personal pronouns (PRP), possessive pronouns (PRP\$), Wh-pronouns (WP) and Wh-possessive pronouns (WP\$)

PRP Personal pronoun

I, you, she, he, it, we, they (as subjects of the sentence)

me, you, her, him, it, us, them (as objects of the sentence)

PRP\$ Possessive pronoun

mine, ours, yours, his, hers, theirs

WP Wh-pronoun

what, who, whom, whoever

WP\$ Possessive Wh-pronoun

whose

WDT Wh-determiner, not a WP Wh-pronoun for CoreNLP

which

Argamon's work

Argamon, working with different groups, has produced impressive evidence on gendered writing style using different types of big data: a large subset of the British National Corpus (BNC) covering a range of different genres (Koppel et al. 2002; Argamon et al. 2003a), a large database of electronic messages of newsgroups postings on different topics (Argamon et al. 2003b), tens of thousands of blogs (Schler et al. 2006; Argamon et al. 2007). The findings are very consistent across methods and data. Male style is characterized by a more frequent use of noun specifiers (determiners – a, the, that, these – numbers – one, two, more, some – modifiers, preposition of, and pronoun its), while female style uses more negations, pronouns – I, you, she, her, their, myself, yourself, herself – prepositions for and with, and conjunction and, auxiliary verbs, and quotations, thus introducing other people's voices. As for bloggers, "male and female bloggers blog about different thing and use different blogging styles." (Schler et al. 2006) Male bloggers talk more about politics, technology, and business than female bloggers, who tend to discuss their personal lives, using a more personal writing style, home, religion, romance, fun. Regardless of gender, however, writing style grows increasingly "male" with age.

Pennebaker's work

Pennebaker's work shows how function words are associated to style and how these vary across age and gender. "The ways people use function words reflects their linguistic style" (Chung and Pennebaker 2007: 347). Using a data archive of some 100,000 text files of different kinds of writing of some 80,000 people and for nearly 70 million words, Pennebaker shows that **males' writing** contains higher rates of articles and nouns, particularly concrete nouns, while **females** use more verbs (especially auxiliary verbs) and first person singular pronouns (Chung and Pennebaker 2007: 353-354). **Age** also affects style, older people displaying greater cognitive complexity and, "interestingly," using "more future tense and less past tense the older they get" (Chung and Pennebaker 2007: 354). But Pennebaker extends the analysis of function words beyond demographics, arguing that **function words "carry an array of psychological meanings"** (Chung and Pennebaker 2007: 355; See also Pennebaker and King 1999) and where "pronouns are among the most revealing" since they "may provide insight into people's level of social integration as well as self-focus." (Pennebaker et al. 2003: 570, 569) Depressed people use

more “I” references than non-depressed people. “Too much attention to the self [through the use of “I”] is associated with highly negative emotional states such as depression,” as shown even by the writing of suicidal poets (Chung and Pennebaker 2007: 352, 351) Yarkoni (2012), on the basis of a small sample of 694 blogs and the LIWC software, also shows the high correlation between personality and language use, choice of words clearly associated to the Big Five factors of personality: neuroticism, extraversion, openness, agreeableness, and conscientiousness.

Do your documents show a differential use of function words?

Does the use of function words vary along a text, if you plot function words by sentence index?

For more information on tag values, see the TIPS files on Part of Speech Tags (POSTAG) and Dependency Relations (DEPREL)

References

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